A pure product fills every basis pairing

coefficient	$v_1 = 3$	$v_2 = 4$
$u_1 = 1$	$1 \times 3 = 3$	$1 \times 4 = 4$
$u_2 = 2$	$2 \times 3 = 6$	$2 \times 4 = 8$